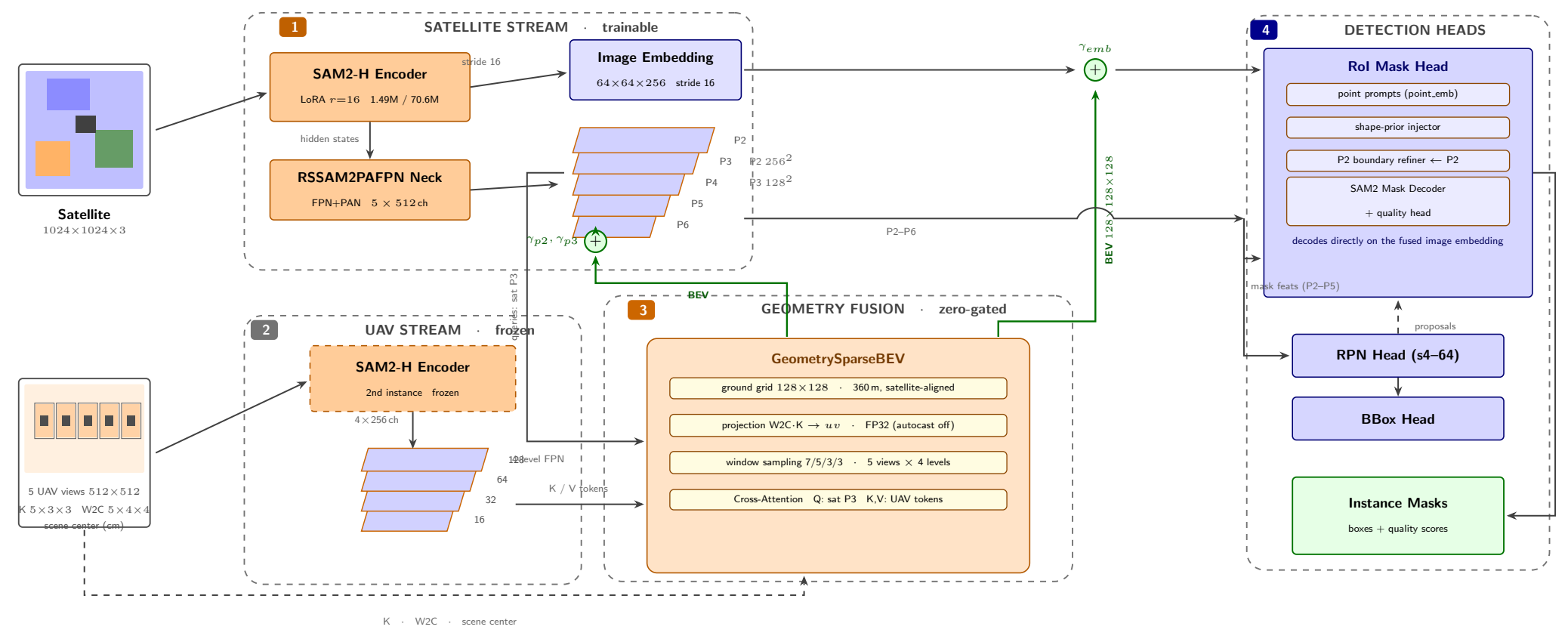

1
SATELLITE STREAM · trainable
SAM2-H Encoder
LoRA r=16 1.49M / 70.6M
stride 16
Image Embedding
64×64×256 stride 16
hidden states
RSSAM2PAFPN Neck
FPN+PAN 5 × 512 ch
P2
P3
P4
P5
P6
P2 256²
P3 128²
P2–P6
Satellite
1024×1024×3
2
UAV STREAM · frozen
SAM2-H Encoder
2nd instance frozen
4×256 ch
128
64
32
16
4-level FPN
K / V tokens
5 UAV views 512×512
K 5×3×3 W2C 5×4×4
scene center (cm)
3
GEOMETRY FUSION · zero-gated
BEV
GeometrySparseBEV
ground grid 128×128 · 360 m, satellite-aligned
projection W2C·K → uv · FP32 (autocast off)
window sampling 7/5/3/3 · 5 views × 4 levels
Cross-Attention Q: sat P3 K,V: UAV tokens
queries: sat P3
BEV 128×128×128
4
DETECTION HEADS
RoI Mask Head
point prompts (point_emb)
shape-prior injector
P2 boundary refiner ← P2
SAM2 Mask Decoder
+ quality head
decodes directly on the fused image embedding
mask feats (P2–P5)
proposals
RPN Head (s4–64)
BBox Head
Instance Masks
boxes + quality scores
K · W2C · scene center